

Project #2 - Topology Optimization

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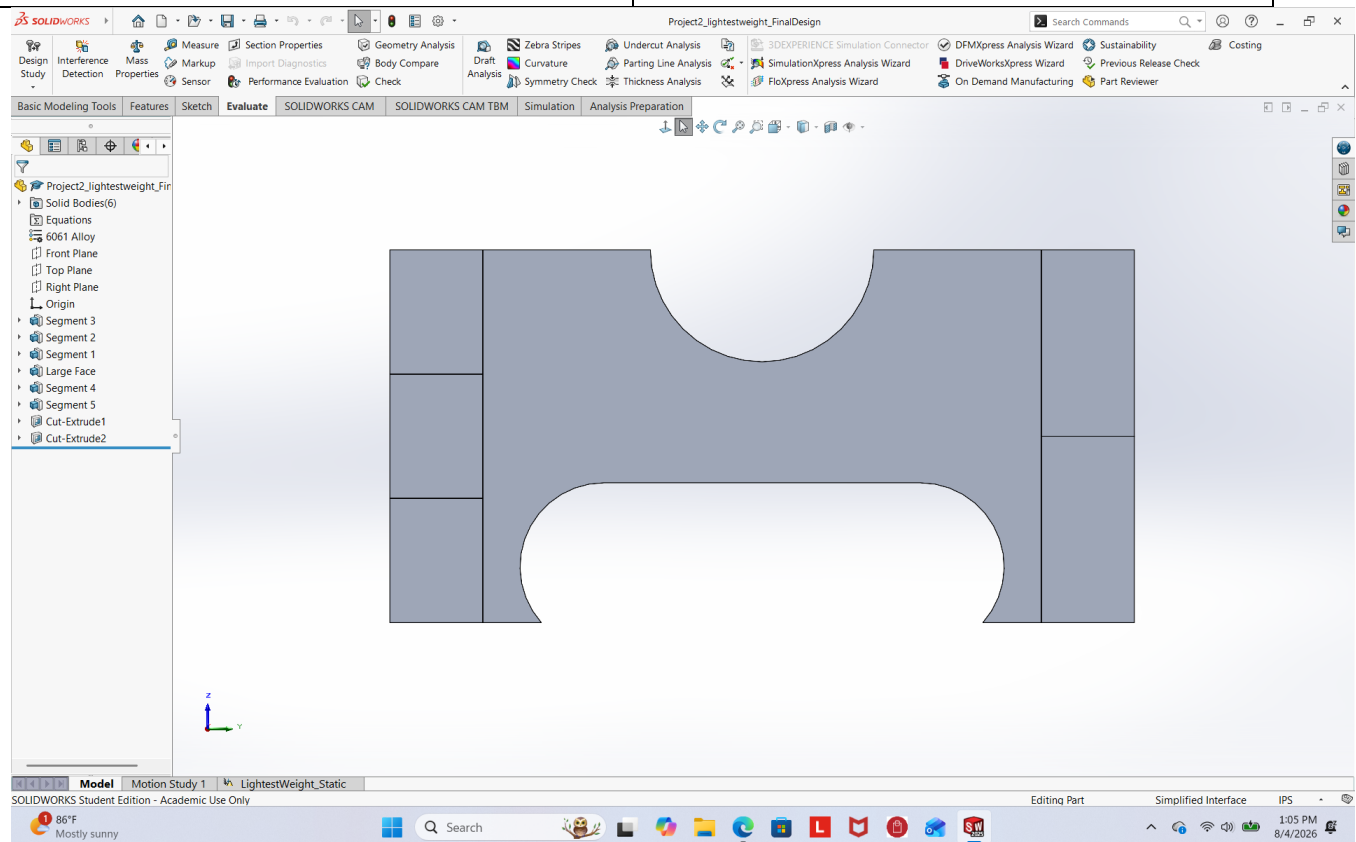


Figure 1. Overall picture of the final optimized lightest-weight part shape.

Overview

This project used SolidWorks Simulation to optimize a 1.000 in by 2.000 in by 0.125 in 6061 aluminum part subjected to a distributed 50.000 lbf load while the opposite end was fixed. Two design approaches were investigated. Approach 1 minimized a cost function that balanced normalized mass and resultant displacement, while Approach 2 pursued a lightest-weight final geometry that remained below the 1.00000 mil resultant-displacement limit and retained a safe stress margin. Five cost iterations were evaluated, the lowest-cost design was verified with a nine-point mesh convergence study, and the lightest-weight design was compared with a manually constructed four-element one-dimensional nodal model.

Optimization and Cost Discussion

Starting Blank

The starting blank retained the original 0.600 in diameter semicircular top cutout and the fixed and loaded end faces specified for the assignment. The SolidWorks study used 6061 Alloy, bonded interactions between the six solid bodies, a fixed restraint on the three left faces, and a distributed 50.000 lbf force on the three right faces. The blank had a mass of 0.022662 lb, a maximum resultant displacement of 0.23631 mil, and a maximum von Mises stress of 1.9996 ksi. The highest stress occurred at the lower edge of the semicircular cutout, which identified the upper neck as an important load path that should not be aggressively removed.

Starting Blank Results

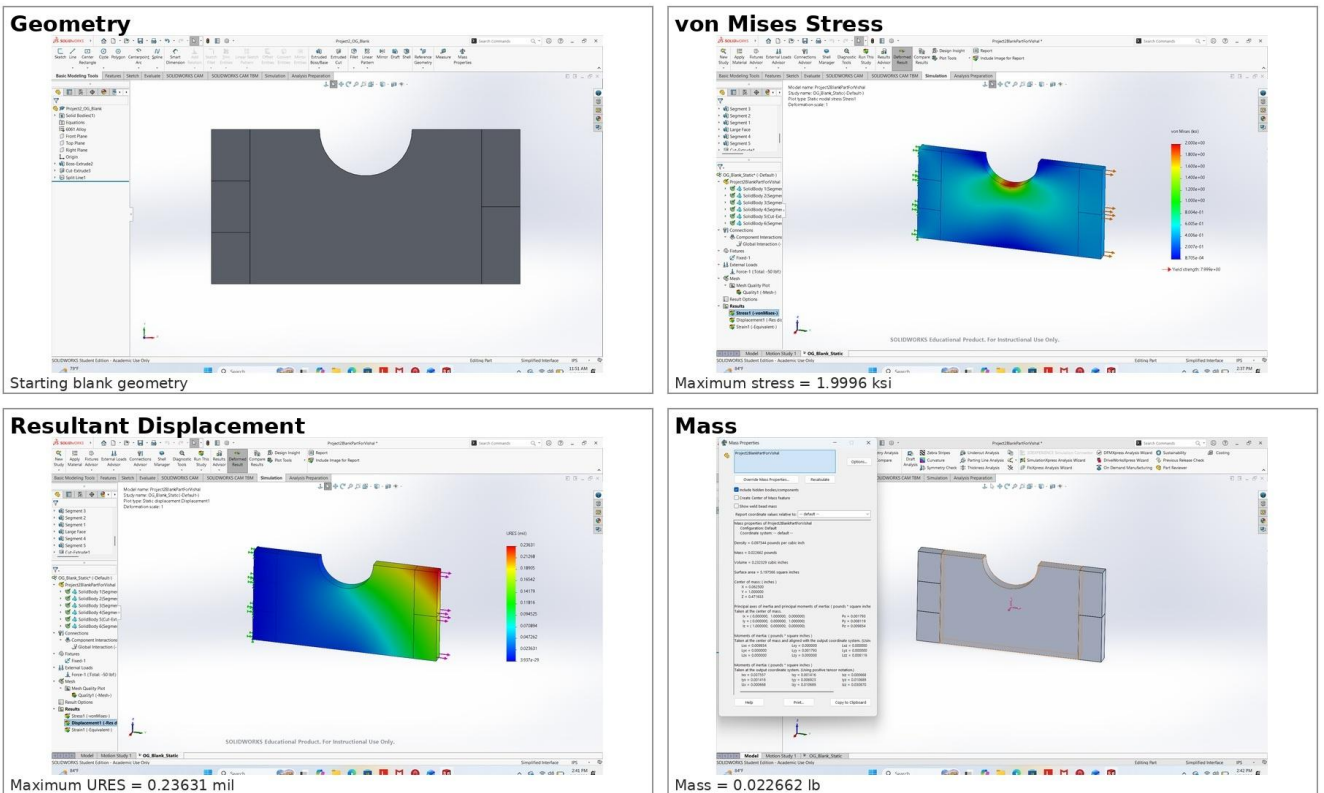


Figure 2. Geometry, stress, resultant displacement, and mass of the starting blank.

Iteration 1

A narrow rounded slot was cut through the lower central portion of the part. This region was selected because the blank stress plot showed comparatively low stress away from the upper semicircular notch and the restrained/loaded interfaces. The cut removed material while maintaining continuous upper and lower load paths. The mass decreased to 0.020816 lb and URES increased only to 0.26504 mil, so the design remained mass-dominant.

Approach 1 - Iteration 1

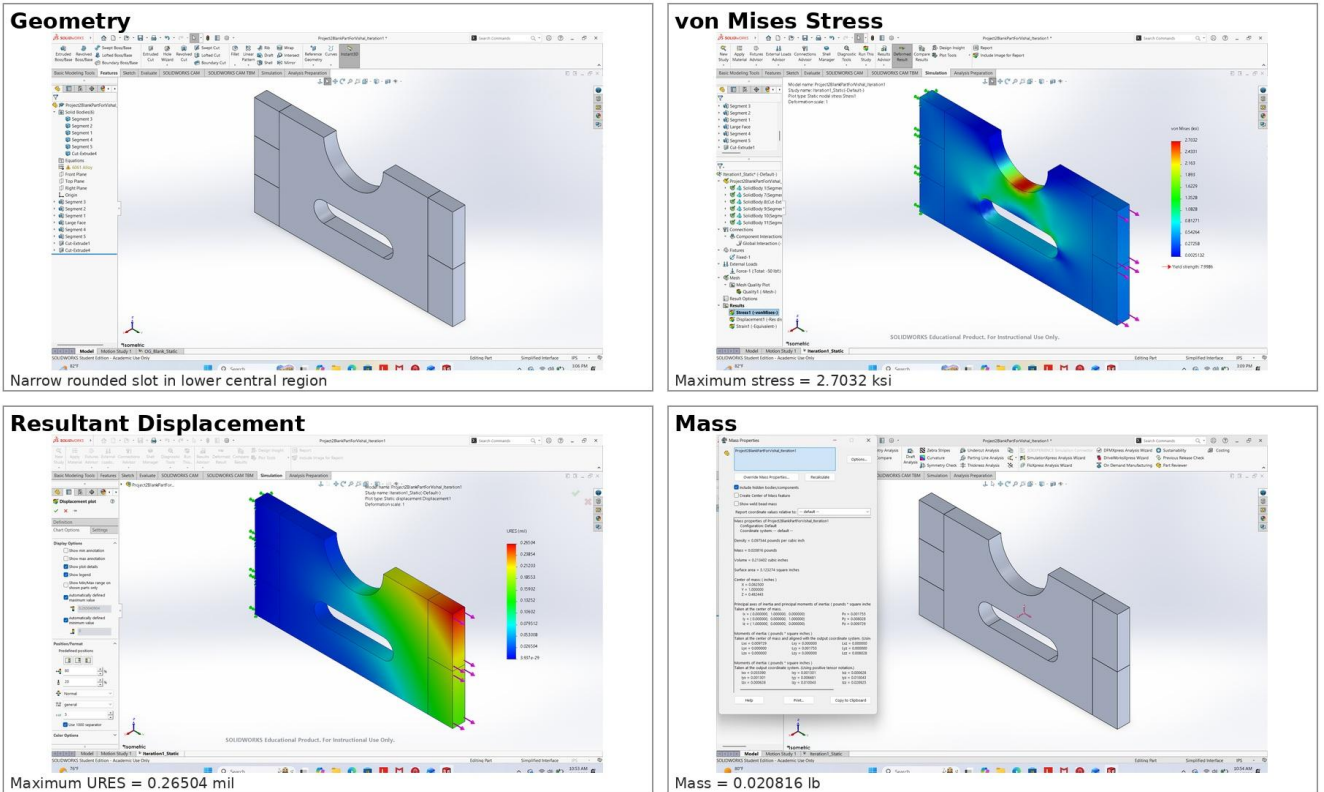


Figure 3. Iteration 1 simulation results.

Iteration 2 - Lowest-Cost Design

The internal slot was enlarged into a broad rounded opening to remove additional low-stress material. Material was retained around the semicircular notch, at both end faces, and along the lower ligament to preserve connectivity. The mass decreased to 0.014400 lb while URES remained 0.34727 mil. This combination produced the minimum total cost of all documented cost iterations, so Iteration 2 was selected for the mesh-convergence study.

Approach 1 - Iteration 2 (Lowest Cost)

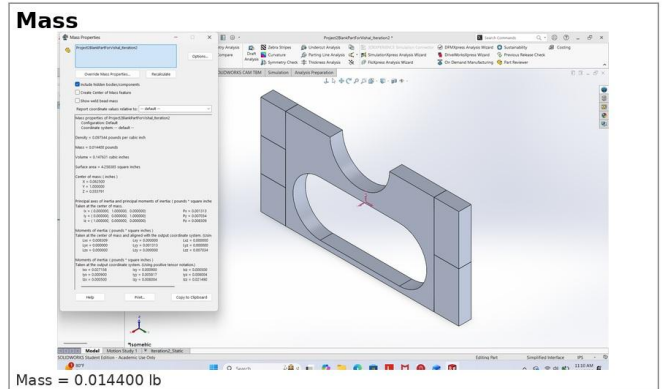
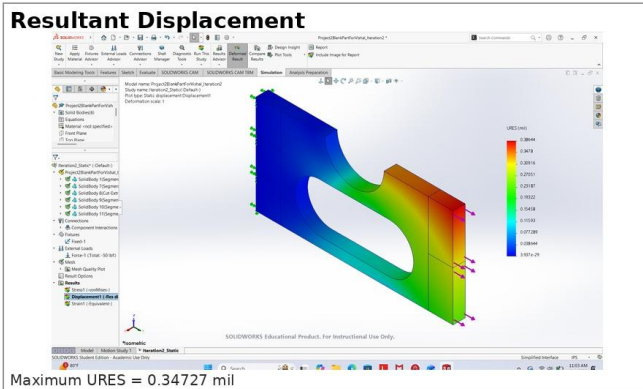
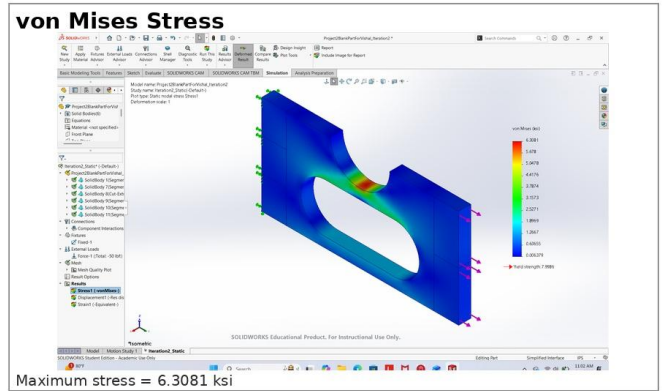
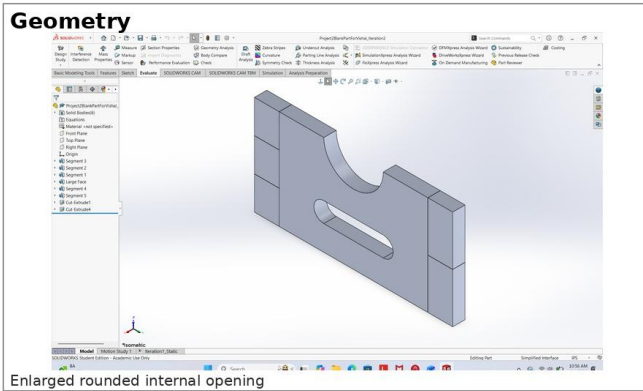


Figure 4. Iteration 2 simulation results - lowest-cost design.

Iteration 3

The opening was expanded again and the central ligaments were made thinner to force a transition toward displacement-dominant behavior. The mass decreased to 0.013661 lb, but URES increased to 0.60330 mil. The stress concentration at the narrow upper neck also increased substantially. This result demonstrated that additional removal in the central load path provided only a small mass reduction while causing a much larger stiffness penalty.

Approach 1 - Iteration 3

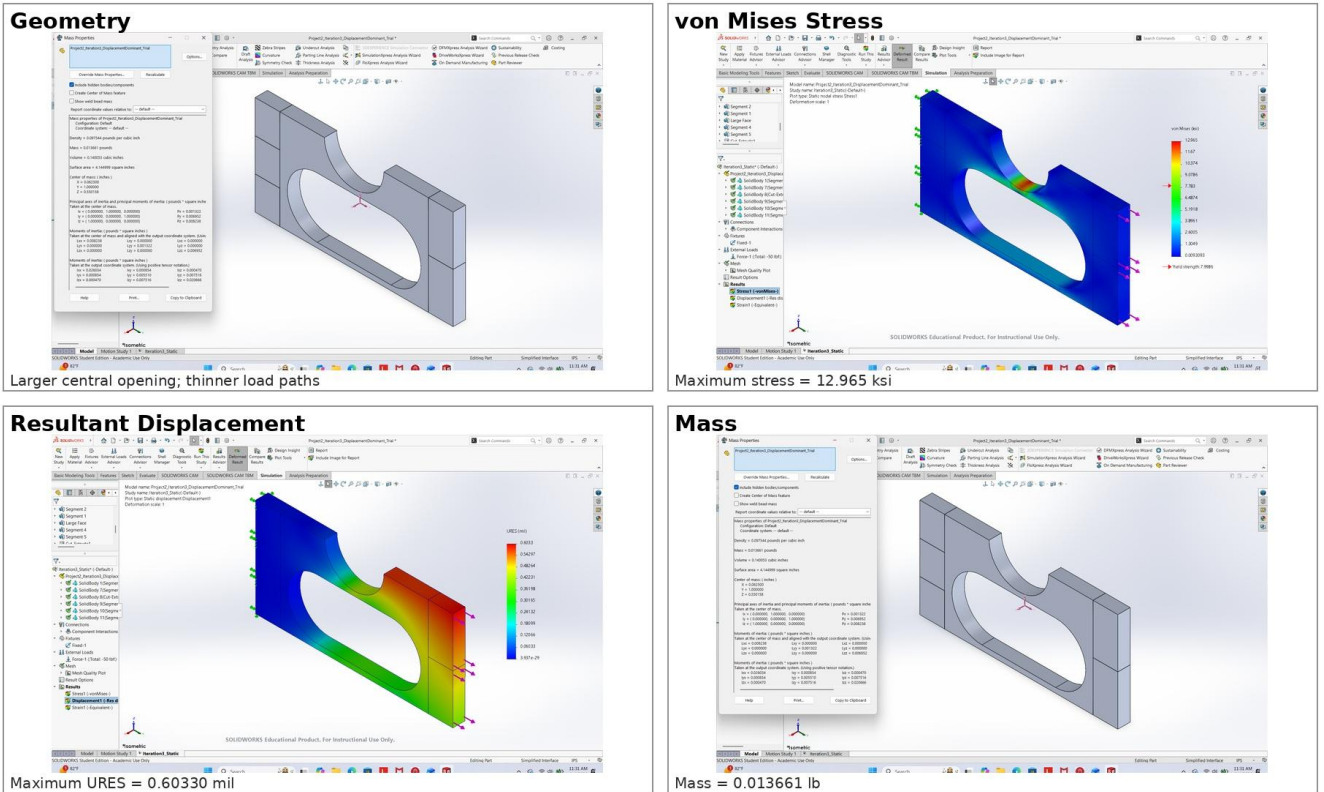


Figure 5. Iteration 3 simulation results.

Iteration 4

Material was restored around the critical neck and along the perimeter of the internal opening. This increased the mass slightly to 0.013995 lb but reduced URES to 0.40305 mil. The normalized mass term again became larger than the displacement term, creating the required transition back to mass-dominant behavior and confirming that reinforcement near the notch was effective.

Approach 1 - Iteration 4

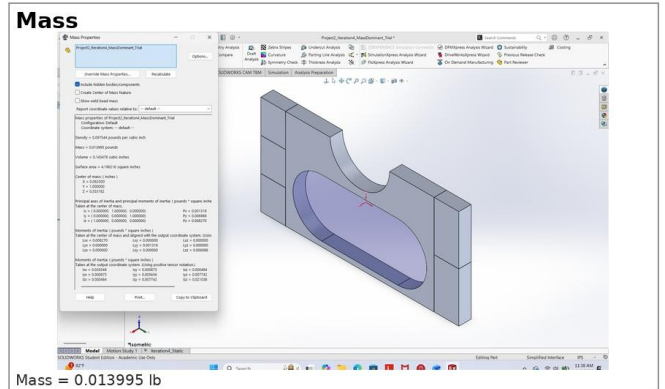
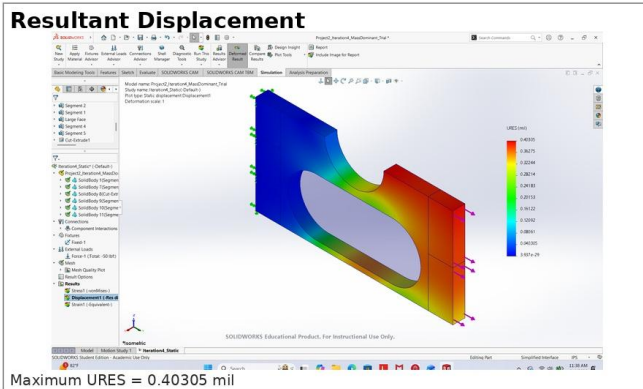
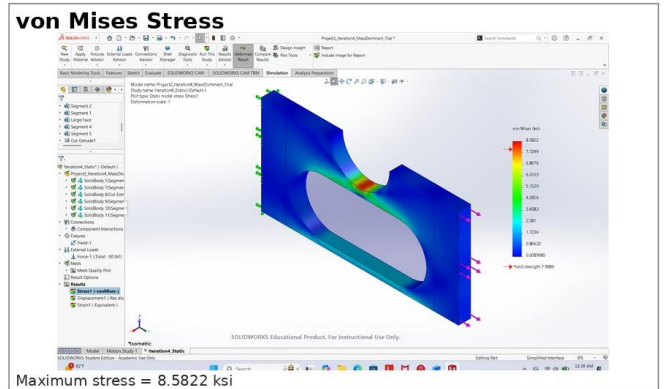
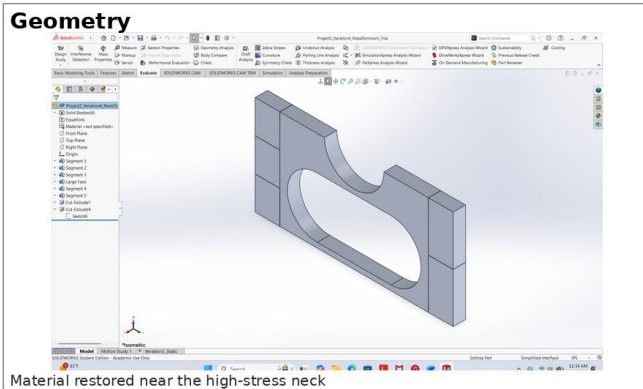


Figure 6. Iteration 4 simulation results.

Iteration 5

A final cost iteration removed material from the interior opening while retaining the reinforced end regions. The mass decreased to 0.013595 lb and URES increased to 0.67174 mil. The displacement term became dominant again, creating the third documented dominant-variable transition. The stress plot also showed that the narrow upper load path was increasingly sensitive to further removal.

Approach 1 - Iteration 5

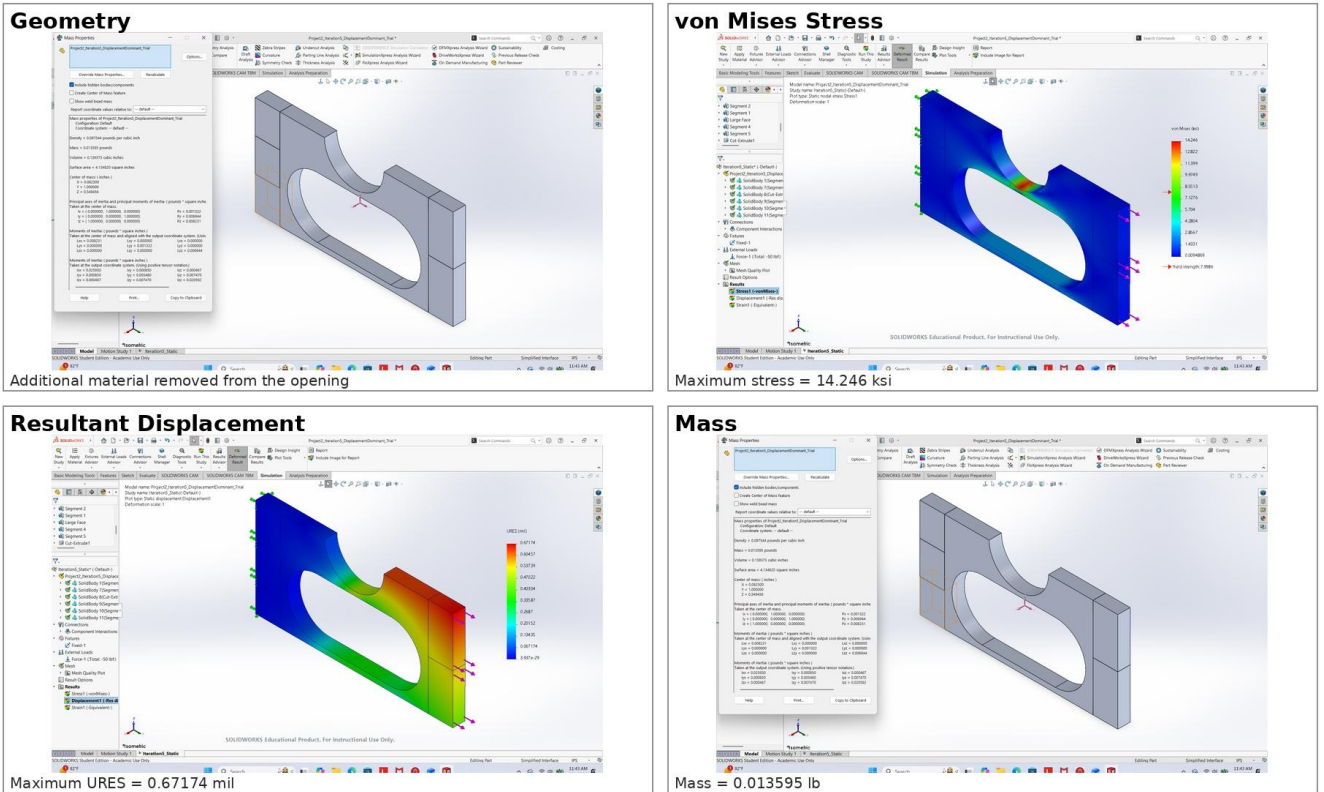


Figure 7. Iteration 5 simulation results.

Approach 2 - Lightest-Weight Design

The final Approach 2 geometry used a large open lower profile with curved transitions and retained material through the upper central bridge and both end interfaces. The selected design had a mass of 0.017049 lb, a maximum resultant displacement of 0.94237 mil, and a maximum von Mises stress of 2.8051 ksi. The displacement remained below the required 1.00000 mil limit, and the peak stress was below the 6061 Alloy yield strength shown in the study. Although several Approach 1 trial geometries had lower raw mass, their thin necks produced substantially higher peak stresses; therefore, the selected Approach 2 geometry was used as the lightest practical final design for validation.

Approach 2 - Selected Lightest-Weight Design

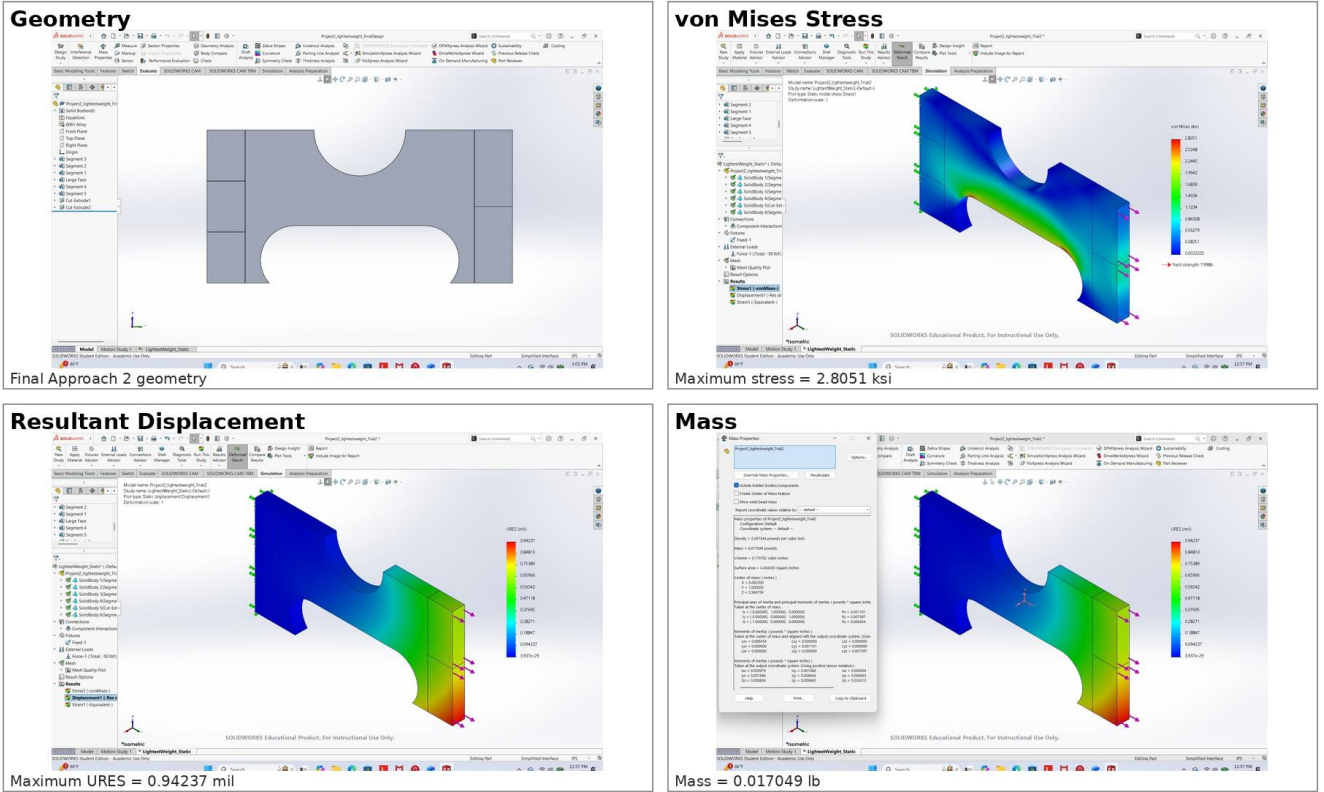


Figure 8. Selected lightest-weight design simulation results.

Individual Cost Table

The cost function combined the normalized mass and normalized resultant displacement with equal weighting. The maximum values were the blank mass, 0.022662 lb, and the allowable displacement, 1.00000 mil.

$$Cost = \sqrt{[(mass / 0.022662)^2 + (URES / 1.00000)^2] / 2}$$

Design	Mass (lb)	URES (mil)	Mass Cost	Displacement Cost	Total Cost	Dominant Variable
Blank	0.022662	0.23631	1.000000	0.236310	0.726582	Mass
Iteration 1	0.020816	0.26504	0.918542	0.265040	0.676005	Mass
Iteration 2	0.014400	0.34727	0.635425	0.347270	0.512036	Mass
Iteration 3	0.013661	0.60330	0.602815	0.603300	0.603058	Displacement
Iteration 4	0.013995	0.40305	0.617554	0.403050	0.521451	Mass
Iteration 5	0.013595	0.67174	0.599903	0.671740	0.636835	Displacement
Lightest Weight	0.017049	0.94237	0.752317	0.942370	0.852655	Displacement

Table 1. Cost of the five Approach 1 iterations and the Approach 2 lightest-weight design.

Iteration 2 was the lowest-cost design with a total cost of 0.512036. The dominant-variable sequence for the five iterations was Mass, Mass, Displacement, Mass, and Displacement. The transitions from Iteration 2 to 3, Iteration 3 to 4, and Iteration 4 to 5 satisfied the requirement for at least three changes in the dominant parameter. The Approach 2 design had a higher cost because it intentionally used more material to maintain a lower stress level while operating close to the displacement limit.

Mesh Size Analysis

A mesh-size convergence study was conducted on Iteration 2 because it had the lowest calculated cost. Nine global element sizes from 0.12000 in to 0.020000 in were evaluated. The mesh tolerance was kept equal to 5% of the corresponding element size. The maximum resultant displacement increased from 0.34315 mil for the coarsest mesh to 0.34693 mil for the finest mesh, a total difference of approximately 1.0896%. After an element size of about 0.080000 in, the displacement remained within approximately 0.00075 mil of the final value. The final two results differed by only 0.05477%, which demonstrates adequate convergence for the design comparison.

Test	Element Size (in)	Tolerance (in)	Maximum URES (mil)
1	0.12	0.006	0.34315
2	0.1	0.005	0.34386
3	0.08	0.004	0.34735
4	0.07	0.0035	0.34749
5	0.0615009	0.00307505	0.34727
6	0.05	0.0025	0.34723
7	0.04	0.002	0.34686
8	0.03	0.0015	0.34674
9	0.02	0.001	0.34693

Table 2. Mesh convergence data for the lowest-cost design.

Mesh Convergence Study - Iteration 2

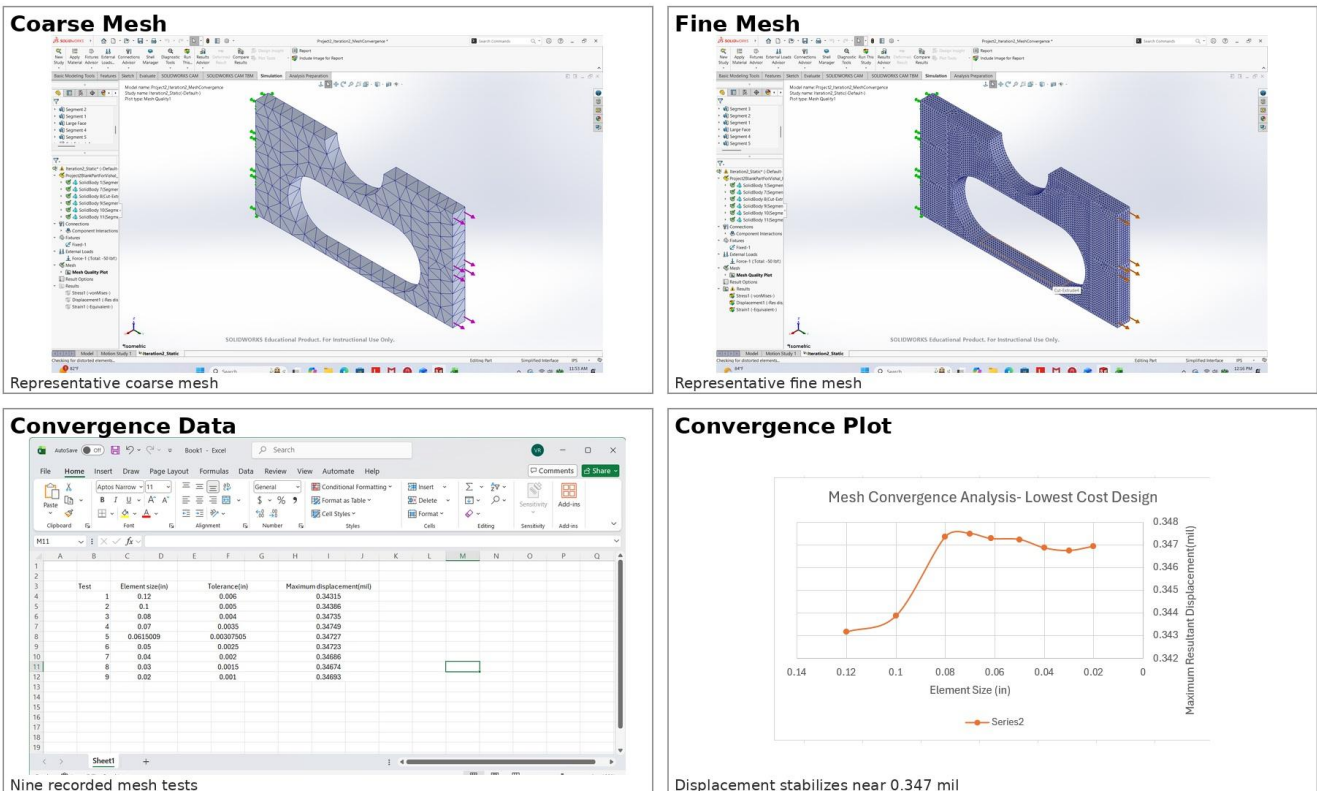


Figure 9. Coarse mesh, fine mesh, convergence data, and convergence plot for Iteration 2.

Nodal Analysis

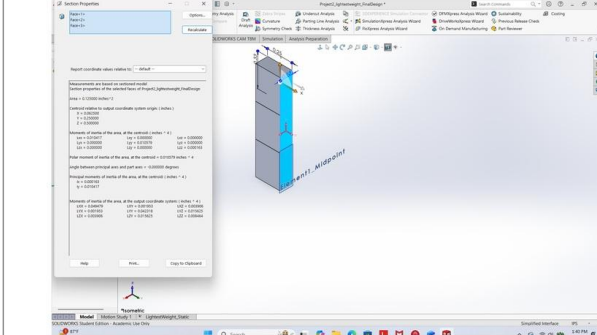
A four-element, five-node one-dimensional axial model was used to estimate displacement parallel to the 2.0000 in long axis of the lightest-weight design. The nodes were placed at 0.0000, 0.50000, 1.0000, 1.5000, and 2.0000 in so that all four elements had a length of 0.50000 in. Cross-sectional areas were evaluated at the midpoint of each element using SolidWorks section properties. The model used the 6061 Alloy elastic modulus $E = 10,007,603.9$ psi and the measured part thickness of 0.12500 in.

Element	Nodes	Midpoint (in)	Length (in)	Area (in ²)
1	1-2	0.25000	0.50000	0.125000
2	2-3	0.75000	0.50000	0.057396
3	3-4	1.25000	0.50000	0.057396
4	4-5	1.75000	0.50000	0.125000

Table 3. Four-element nodal-model geometry and cross-sectional areas.

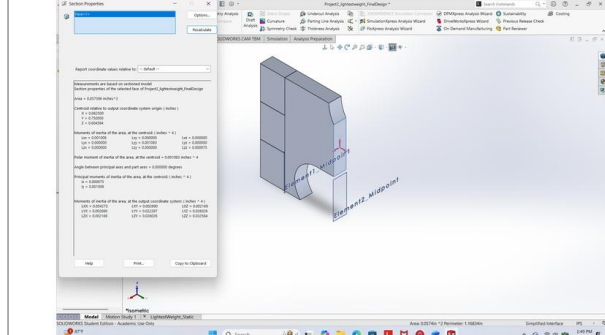
Cross-Sectional Areas Used in the Four-Element Nodal Model

Element 1



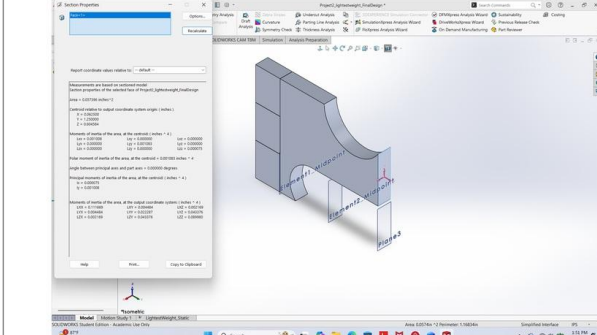
A1 = 0.12500 in² at x = 0.250 in

Element 2



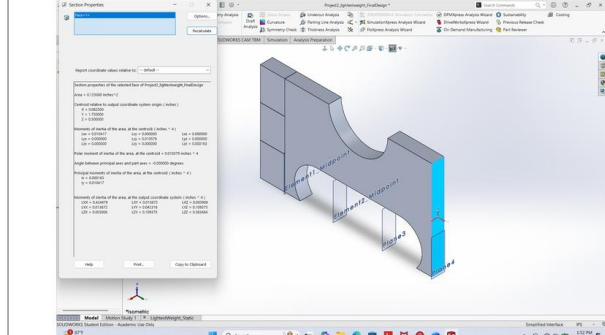
A2 = 0.057396 in² at x = 0.750 in

Element 3



A3 = 0.057396 in² at x = 1.250 in

Element 4



A4 = 0.12500 in² at x = 1.750 in

Figure 10. SolidWorks section-property measurements used to approximate the four element areas.

Element Stiffness

$$k_j = EA_j/L_j$$

Element Stiffness	Value (lbf/in)
k1	2,501,900.97500
k2	1,148,792.86689
k3	1,148,792.86689
k4	2,501,900.97500

Table 4. Calculated axial stiffness of each element.

Global Stiffness Equations

The left node was fixed, so $u_1 = 0$. The remaining displacement unknowns were solved from the reduced global stiffness system shown below. The 50.000 lbf load was applied at Node 5.

3,650,693.84189	-1,148,792.86689	0	0
-1,148,792.86689	2,297,585.73378	-1,148,792.86689	0
0	-1,148,792.86689	3,650,693.84189	-2,501,900.97500
0	0	-2,501,900.97500	2,501,900.97500

$$[K]\{u\} = \{F\}, \quad \{F\} = [0, 0, 0, 50.000]^T \text{ lbf}$$

Nodal Displacement Solution

Node	Displacement (in)	Displacement (mil)
1	0.000000000000	0.000000
2	0.000019984804	0.019985
3	0.000063508751	0.063509
4	0.000107032698	0.107033
5	0.000127017502	0.127018

Table 5. Four-element nodal-analysis displacement solution.

The calculated loaded-end displacement was 0.127018 mil. Each element carried 50.000 lbf in the idealized series model, and the calculated reaction at the fixed end was 50.000 lbf in the opposite direction. The smaller-area center elements elongated more than the full-area end elements, which is consistent with the axial relation $\delta = FL/(EA)$.

Comparison with SolidWorks

The project specification requires comparison using displacement parallel to the long axis rather than URES. A UY probe of the loaded faces gave an average displacement of 0.22574 mil. The four-element model predicted 0.127018 mil, which is 0.098722 mil lower than the SolidWorks face average, or a 43.733% difference using the SolidWorks value as the reference. The manual model idealizes the part as four straight axial springs and therefore omits bending, shear deformation, local curvature, stress concentrations, and nonuniform displacement across the loaded faces. Those effects explain why the detailed SolidWorks model is more flexible than the simplified one-dimensional calculation.

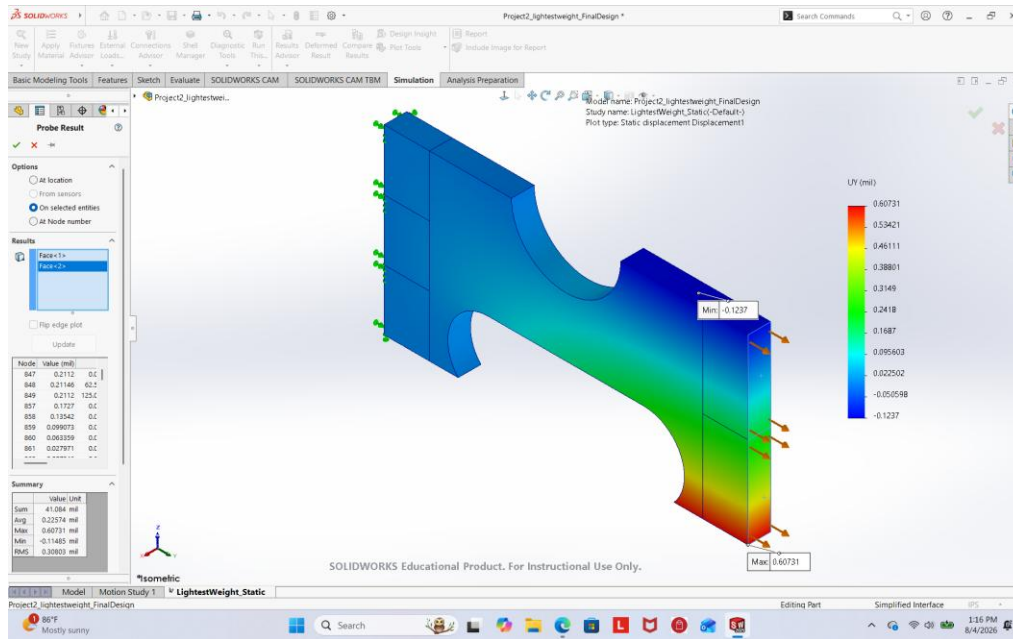


Figure 11. SolidWorks UY probe used for comparison with the one-dimensional nodal model; loaded-face average = 0.22574 mil.

Group Comparison

Group 5 compared each member's lowest-cost design and lightest-weight design. The table below lists the cost of each member's lowest-cost part and the mass of each member's lightest-weight part. All Group 5 members will include the same table.

Group 5 Member #	Name (First and Last)	Cost of Lowest-Cost Part	Mass of Lightest-Weight Part (lb)
1	Connor Bowen Stilwell	0.80568	0.014670
2	Daniel Fisher	0.828575	0.015479
3	Jacob Roberts	0.65872	0.016386
4	Sunny Chittaphong	0.85286	0.014472
5	Vishal Sreevidya Rajan	0.512036	0.017049

Table 6. Group 5 cost and lightest-weight comparison results.

Conclusion

SolidWorks Simulation provided a practical method for relating geometry changes to mass, displacement, and stress. The starting blank had a mass of 0.022662 lb and URES of 0.23631 mil. Iteration 2 produced the lowest cost, 0.512036, with a mass of 0.014400 lb and URES of 0.34727 mil. Its mesh study converged near 0.347 mil, confirming that the design ranking was not controlled by an excessively coarse mesh. The selected lightest-weight design had a mass of 0.017049 lb, URES of 0.94237 mil, and maximum stress of 2.8051 ksi. The four-element nodal analysis predicted 0.127018 mil of long-axis displacement compared with a SolidWorks loaded-face average UY of 0.22574 mil. The difference showed the limitations of a one-dimensional axial approximation for a curved topology-optimized part. The most important design lesson was that low-stress material can be removed effectively, but the narrow load path below the upper semicircular cutout must retain enough material to limit both local stress and global flexibility. The project also demonstrated the importance of mesh convergence, consistent units, and balancing multiple performance objectives instead of minimizing mass alone.

Appendix: 1D Nodal Analysis Calculations

Values shown to 5 or 6 significant figures.

1D Nodal Analysis Calculations

All reported values are shown to 5 or 6 significant figures for the Project 2 lightest-weight design.

Inputs: $E = 1.00076 \times 10^7$ psi $L_e = 0.500000$ in $P = 50$ lbf
 $A_1 = 0.12500$ in² $A_2 = 0.057396$ in² $A_3 = 0.057396$ in² $A_4 = 0.12500$ in²

1. Element Stiffness For each element e : $k_e = A_e E / L_e$ $k_1 = (0.12500 \times 1.00076 \times 10^7) / 0.500000 = 2.5019 \times 10^6$ lbf/in $k_2 = (0.057396 \times 1.00076 \times 10^7) / 0.500000 = 1.14879 \times 10^6$ lbf/in $k_3 = (0.057396 \times 1.00076 \times 10^7) / 0.500000 = 1.14879 \times 10^6$ lbf/in $k_4 = (0.12500 \times 1.00076 \times 10^7) / 0.500000 = 2.5019 \times 10^6$ lbf/in Local element matrix form: $[k]^{(e)} = k_e \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$	2. Global Stiffness Matrix $[K] =$ $\begin{bmatrix} 2.5019 \times 10^6 & -2.5019 \times 10^6 & 0 & 0 & 0 \\ -2.5019 \times 10^6 & 3.65069 \times 10^6 & -1.14879 \times 10^6 & 0 & 0 \\ 0 & -1.14879 \times 10^6 & 2.29759 \times 10^6 & -1.14879 \times 10^6 & 0 \\ 0 & 0 & -1.14879 \times 10^6 & 3.65069 \times 10^6 & -2.5019 \times 10^6 \\ 0 & 0 & 0 & -2.5019 \times 10^6 & 2.5019 \times 10^6 \end{bmatrix}$ $[K]\{u\} = \{F\}$
3. Reduced System After Boundary Conditions Fixed left edge $\Rightarrow u_1 = 0$ Applied end load vector: $\{F_r\} = [0 \ 0 \ 0 \ 50]^T$ lbf $[K_r] =$ $\begin{bmatrix} 3.65069 \times 10^6 & -1.14879 \times 10^6 & 0 & 0 \\ -1.14879 \times 10^6 & 2.29759 \times 10^6 & -1.14879 \times 10^6 & 0 \\ 0 & -1.14879 \times 10^6 & 3.65069 \times 10^6 & -2.5019 \times 10^6 \\ 0 & 0 & -2.5019 \times 10^6 & 2.5019 \times 10^6 \end{bmatrix}$ Solve: $[K_r]\{u_r\} = \{F_r\}$	4. Nodal Displacement Solution $\{u_r\} = [K_r]^{-1}\{F_r\}$ $u_2 = 1.99848 \times 10^{-5}$ in = 0.0199848 mil $u_3 = 6.35088 \times 10^{-5}$ in = 0.0635088 mil $u_4 = 0.000107033$ in = 0.107033 mil $u_5 = 0.000127018$ in = 0.127018 mil Full nodal displacement vector: $\{u\} = [0, 1.99848 \times 10^{-5}, 6.35088 \times 10^{-5}, 0.000107033, 0.000127018]^T$ in Maximum axial displacement in the 1D model occurs at node 5.

These equations and numerical results summarize the procedure used to compute element stiffness, assemble the global stiffness matrix, apply the boundary conditions, and solve for nodal displacements for the 1D hand-calculation model.

Figure A1. 1D nodal analysis calculations used for the hand solution.